multiple of $\frac{h}{2\pi}$ called as reduced Planck's constant – that is, $\hbar$ (read it as h-bar) and the integer n is called as principal quantum number.

$$l = n\hbar \qquad \text{where } \hbar = \frac{h}{2\pi}$$

This condition is known as angular momentum quantization condition.

According to quantum mechanics, particles like electrons have dual nature (Refer unit 7, volume 2 of +2 physics text book). The standing wave pattern of the de Broglie wave associated with orbiting electron in a stable orbit is shown in Figure 9.15.

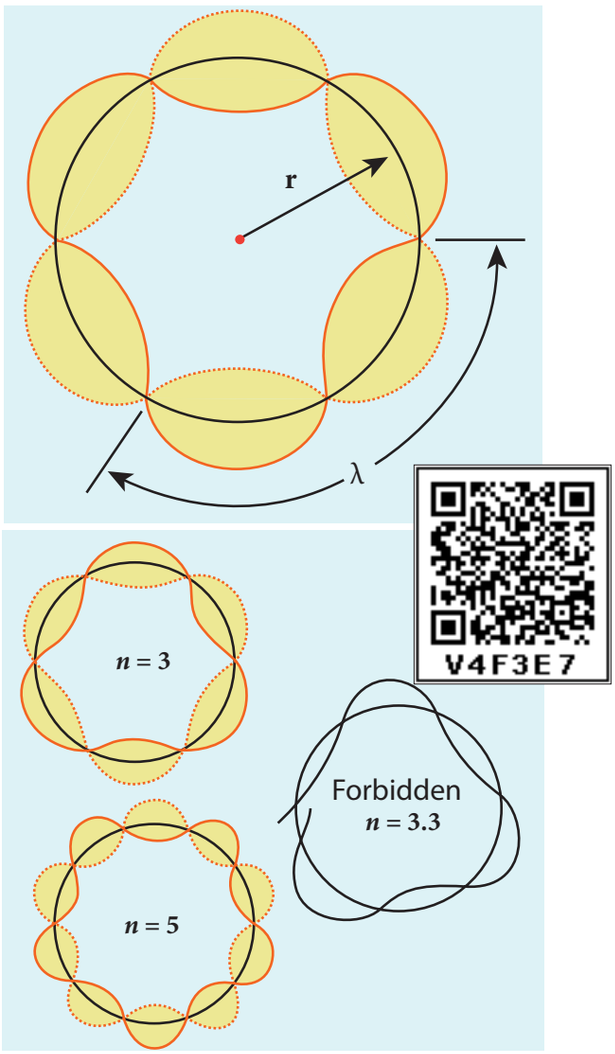


Figure 9.15 Standing wave pattern for electron in a stable orbit

The circumference of an electron's orbit of radius r must be an integral multiple of de Broglie wavelength – that is,

$$2\pi r = n\lambda \qquad (9.14)$$

where n = 1,2,3,.....

But the de Broglie wavelength (λ) associated with an electron of mass m moving with velocity v is $\lambda = \frac{h}{m\upsilon}$ where h is called Planck's constant. Thus from equation (9.14),

$$2\pi r = n\left(\frac{h}{m\upsilon}\right)$$

$$m\upsilon r = n\frac{h}{2\pi}$$

For any particle of mass m undergoing circular motion with radius r and velocity v, the magnitude of angular momentum l is given by

$$l = r(m\upsilon)$$

$$m\upsilon r = l = n\hbar$$

(c) Energy of the electron in orbits is not continuous but only discrete. This is called the quantization of energy. An electron can jump from one orbit to another orbit by absorbing or emitting a photon whose energy is equal to the difference in energy (ΔE) between the two orbital levels (Figure 9.16)

$$\Delta E = E_{final} - E_{initial} = h\nu = h\frac{c}{\lambda}$$

where c is the speed of light and λ is the wavelength and ν is the frequency of the radiation emitted. Thus, the frequency of the radiation emitted is related only to change in atomic energy levels and it does not depend on frequency of orbital motion of the electron.

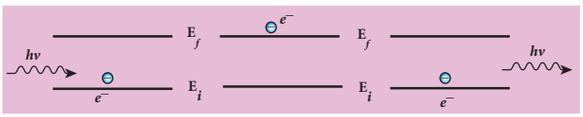


Figure 9.16 Absorption and emission of radiation

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